

CSE 472: Social Media Mining

Project I - Social Media Data Analysis

2025 Palisades Fire on Mastodon

Data collection, information diffusion, user interaction, network measures, and local LLM keyword analysis

DATA	NETWORKS	CONTENT
500 focal posts 200 users	2 network views Gephi layouts	1,500 LLM keywords 500 posts x 3

Submission date: September 2026

Repository: github.com/June0419/CSE472-crawling

Executive Summary

This project examines public Mastodon discussion of the 2025 Palisades Fire. It combines keyword-based post collection, seed-user expansion, two network views, Gephi visualization, structural network measures, and local Llama 3.2 keyword extraction. The final dataset contains 500 focal hashtag posts, 431 context posts retained for reply chains, and 200 relevant users.

The information-diffusion graph is highly fragmented (470 weak components; density 0.000532), while the user network has a large component containing 62.5% of users. Its average degree is 1.70. PageRank, degree, and betweenness distributions all show concentration around a small group of structurally prominent accounts. A local LLM generated exactly three keywords for each focal post, producing a 1,500-keyword corpus for the final word cloud.

1. Disaster Event and Research Objective

The Palisades Fire began on January 7, 2025, southeast of Palisades Drive in Pacific Palisades, Los Angeles County. CAL FIRE's incident record reports 23,448 acres burned, 6,845 structures destroyed, 975 damaged, and 12 confirmed civilian fatalities [1]. The event generated immediate safety information and longer-running discussion about evacuation, losses, recovery, insurance, policy, and secondary hazards.

The objective is to study how disaster-related information appears and propagates on Mastodon, identify structurally important posts and users, and summarize the content themes with an LLM. Results describe the collected public sample rather than all Mastodon users or the wider population.

2. Data Collection

2.1 Keyword-based posts

The collector queried the Mastodon REST API [2] on mastodon.social using three event-specific hashtags: **#PalisadesFire**, **#PacificPalisadesFire**, and **#PalisadesWildfire**. These terms combine the event name, affected location, and disaster type, reducing ambiguity while covering common naming variants. Posts were deduplicated by status ID. Reply-context requests added parent and descendant statuses needed to reconstruct conversation edges.

Collection item	Count / value
Focal hashtag posts	500
Context posts for network edges	431
Total statuses stored	931
Source instance	https://mastodon.social
Collected at (UTC)	2026-09-26T03:59:07.890164+00:00

2.2 User-based expansion

Six seed accounts were selected after verifying that their collected posts addressed the Palisades Fire. They include active disaster-information accounts and journalists. The crawler expanded through mentions, replies, and reblogs, then filled remaining slots with authors of verified event posts. This produced 200 users and 170 unique undirected interaction edges.

Seed users: @ai6yr@m.ai6yr.org, @ikluft@avgeek.social, @kellyromanych, @CA_Politics_Huddle, @hollywoodreporter, @deadline

2.3 Stored attributes and ethics

The two required JSON datasets retain public post identifiers, timestamps, language, cleaned text, reply/reblog links, hashtags, engagement counts, account metadata, and interaction types. The project uses only public API responses, keeps the access token outside Git, and reports aggregate patterns. Usernames appear only where needed to interpret network prominence; centrality is not treated as credibility.

3. Network Construction and Gephi Visualization

Python and NetworkX [3] were used to construct both graphs and export GEXF/GraphML files. Gephi [4] provided the final layouts and styling. Node size is proportional to degree in both views. The project files retain every node; only isolated nodes are hidden from the report PNGs to improve legibility.

Network	Nodes	Edges	Direction	Edge meaning
Information diffusion	931	461	Directed	Parent/original post -> reply or reblog
User interaction	200	170	Undirected	Mention, reply, or reblog interaction

3.1 Information diffusion network

Each node is a post or reply. A directed edge runs from a parent/original status to a collected reply or reblog. ForceAtlas2 was used so related cascades attract one another. Nodes are colored by post type and sized by degree.

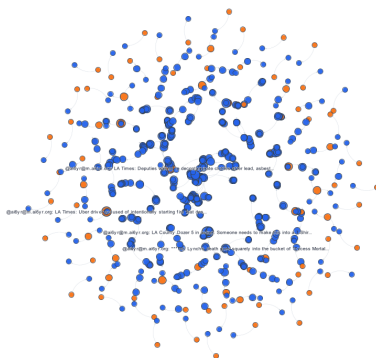


Figure 1. Information diffusion network in Gephi (ForceAtlas2; node size = degree; isolates hidden in the report image).

3.2 User interaction network

Each node is a sampled account. An undirected weighted edge records at least one mention, reply, or reblog interaction; edge weight counts repeated interactions. Yifan Hu was used as a deliberately different layout. Node color represents Louvain community and node size represents degree.

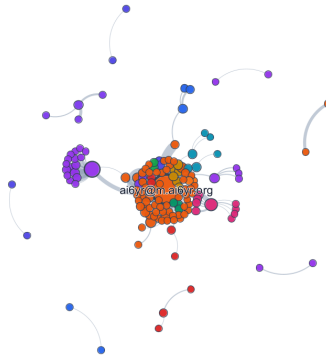


Figure 2. User interaction network in Gephi (Yifan Hu; node size = degree; color = modularity community; isolates hidden in the report image).

4. User Network Measures

Three meaningful measures were selected: PageRank for recursive structural influence, degree for immediate 1-hop connectivity, and betweenness centrality for brokerage across shortest paths. The complete per-user values are in `user_node_measures.csv`.

Metric	Result
Average 1-hop relations (global mean degree)	1.70
Density	0.008543
Connected components	65
Isolates	56
Largest component	125 users (62.5%)
Average clustering coefficient	0.1918
Louvain modularity	0.4417

4.1 PageRank

The distribution is strongly right-skewed. Most users receive small PageRank scores, while a few hub accounts receive much larger values. This indicates concentrated structural influence rather than an evenly distributed interaction network.

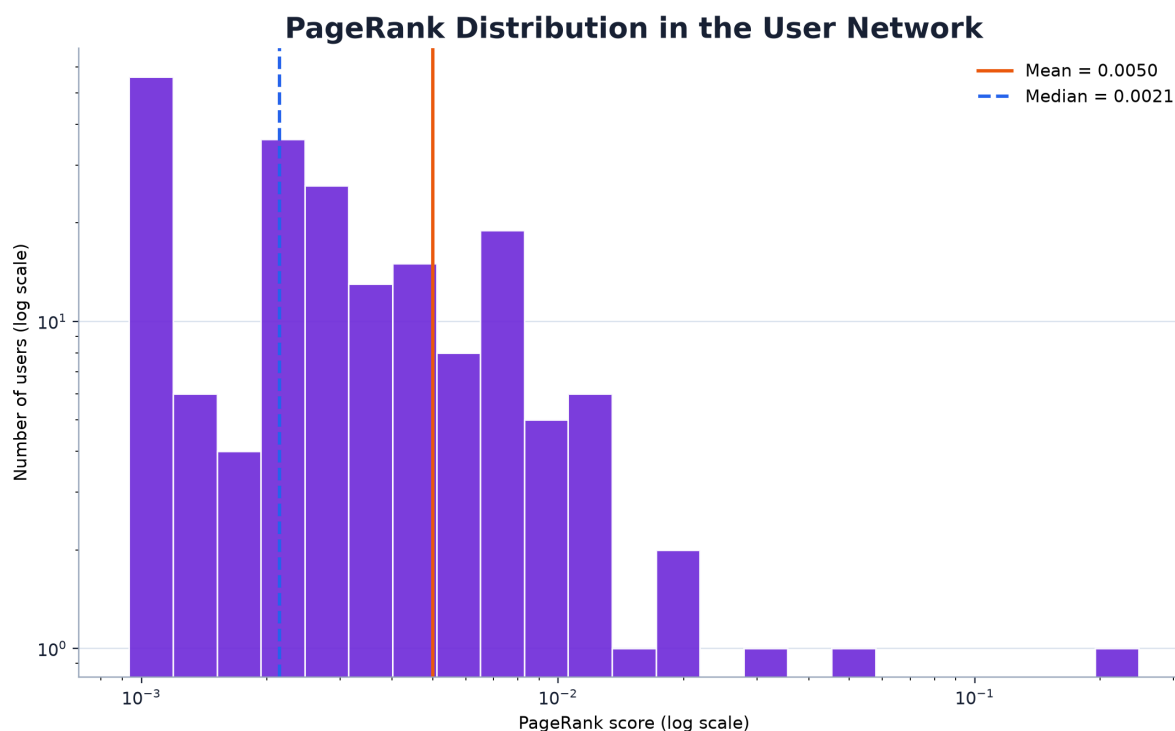


Figure 3. PageRank histogram. Log scales reveal the long upper tail while retaining the dense low-score region.

4.2 Degree and local/global connectivity

At the local level, each node's degree is its number of distinct immediate interaction partners. At the global level, the average is 1.70 relations per user. The coexistence of 56 isolates with a 91-degree hub shows why the mean alone is not representative of the typical user.

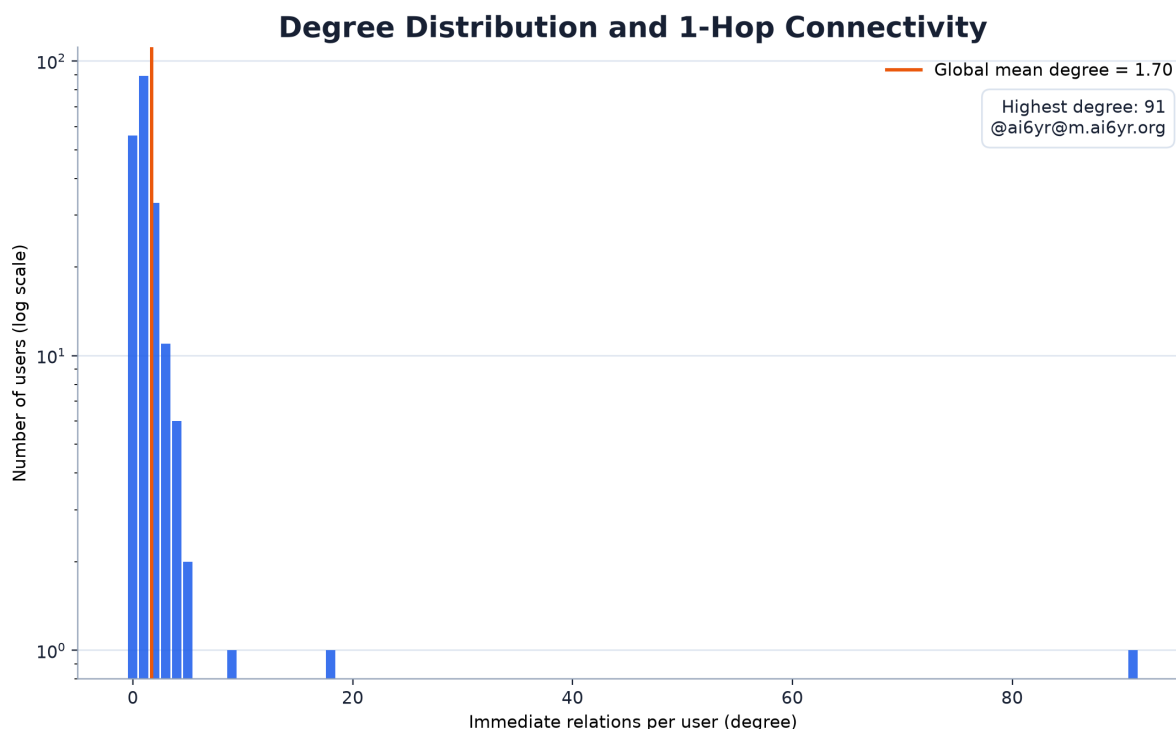


Figure 4. Degree distribution and the required global mean of 1-hop relations.

4.3 Betweenness centrality

Betweenness is zero for many users because they do not lie on a shortest path between other sampled accounts. The positive scores also form a long tail, so a small set of accounts bridges otherwise separated regions of the interaction graph.

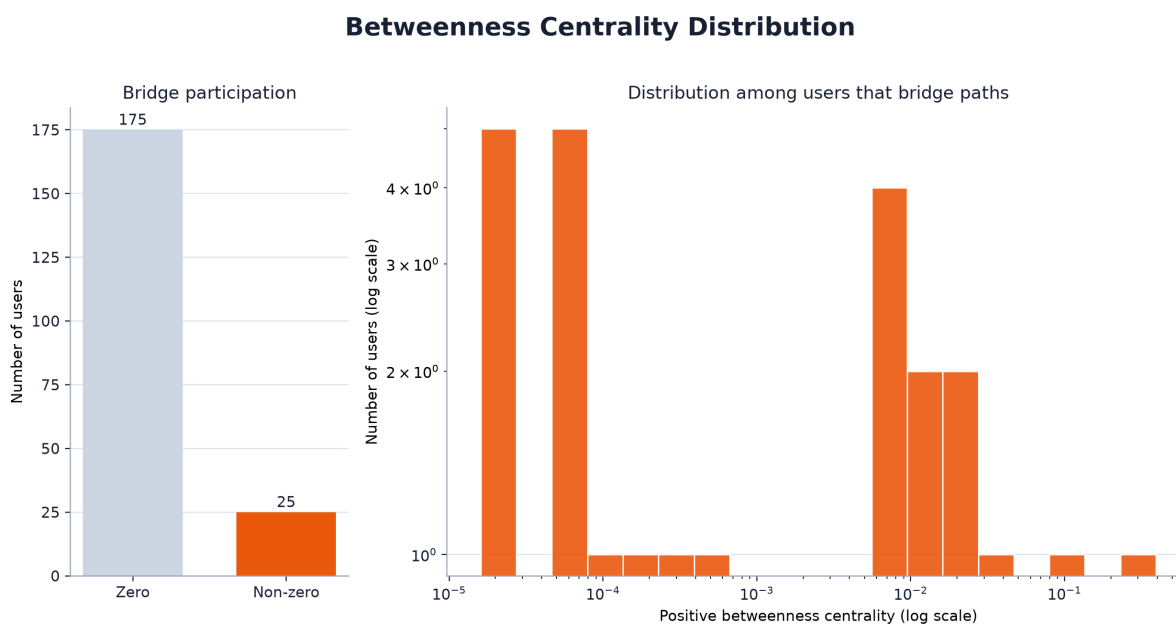


Figure 5. Betweenness distribution, separating zero values from the positive-score histogram.

4.4 Structurally prominent accounts

Account	Degree	Weighted degree	PageRank	Betweenness
@ai6yr@m.ai6yr.org	91	405	0.23582	0.37362
@ikluft@avgeek.social	18	91	0.05141	0.09350
@BakerRL75@m.ai6yr.org	9	26	0.01943	0.02772
@exador23@m.ai6yr.org	5	22	0.01301	0.00640
@jeridansky@sfb.social	5	38	0.01808	0.00009

The rankings show prominence inside this collected sample. They should not be interpreted as measures of authority, truthfulness, or influence outside the selected hashtags and seed expansion.

5. LLM Content Analysis

Llama 3.2 3B was run locally through Ollama. Structured JSON output [5] and temperature 0 were used to require exactly three distinct English keywords for each of the 500 focal hashtag posts. This yielded 1,500 keyword assignments. Local inference avoided sending public post text to a hosted LLM API. Llama 3.2 is an instruction-tuned text model intended for tasks including summarization [6].

5.1 Exact prompt

You are a precise content-analysis assistant for a disaster social-media study.
For every Mastodon post, return exactly three distinct, concise English keywords or short phrases that are explicitly supported by the post.

Prioritize disaster-related named events, affected places, hazards, impacts, emergency response, recovery, policy, and community actions. Use 1-3 words per keyword. Expand compressed hashtags and never split a proper name across separate keywords: #PalisadesFire becomes "palisades fire", #EatonFire becomes "eaton fire", and #LosAngeles becomes "los angeles". Do not return isolated fragments such as "los", "angeles", "palisades", "eaton", or the generic word "fire" when the post states a more specific event or place.

Example: a post saying "LA County sues State Farm over wildfire claims #EatonFire #PalisadesFire #insurance" should return ["eaton fire", "palisades fire", "insurance claims"]. A post saying "Palisades Fire burns in Los Angeles during a windstorm" should return ["palisades fire", "los angeles", "windstorm"].

Normalize obvious variants to a consistent form. Ignore URLs, account names, platform terms, engagement requests, timestamps, and non-topical boilerplate. Do not infer sentiment, causes, or facts that the post does not state. Return only data matching the supplied JSON schema.

5.2 Example outputs

Post excerpt	Three LLM keywords
Historic Malibu Winery Closes After Losing Tasting Room to Palisades Fire #Lifestyle #LifestyleNews #Malibu #PalisadesFire #Wine #Winesandspirits	malibu winery; palisades fire; wines and spirits
Via Supervisor Barger: "Fire survivors, in partnership with a UCLA research team, are building the nation's first public record of which insurers paid what they owed, which left fa	eaton fire; palisades fire; insurer claims
LA Times: L.A. County sues State Farm over its handling of wildfire claims #wildfires #EatonFire #PalisadesFire #insurance #disaster	la county; palisades fire; state farm
91 years ago, we lost #WillRogers. 16 years ago, I made a real bucket-list trip to the 1935 crash site near distant Barrow, Alaska. Despite Will's home loss in the #Palisadesfire I	will rogers; palisades fire; alaska
Billionaire Rick Caruso Brings Back Palisades Village After Wildfire: "This Is One Chapter of the Comeback Story" #Business #BusinessNews #Pacificalisades #PalisadesFire #Palisade	palisades village; palisades fire; rick caruso

5.3 Keyword frequency and word cloud

2025 Palisades Fire: Llama 3.2 Keyword Cloud

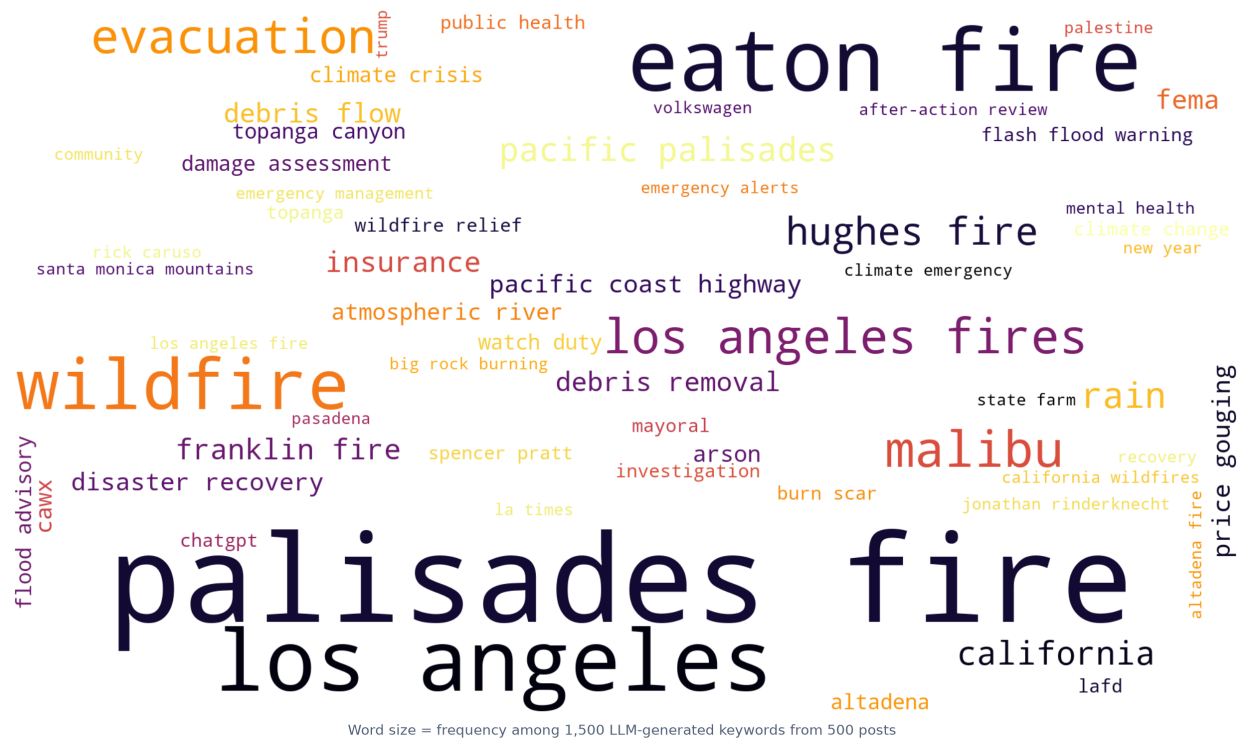


Figure 6. Word cloud generated from the 1,500 LLM keywords. Word size is proportional to frequency.

Most Frequent LLM-Generated Keywords

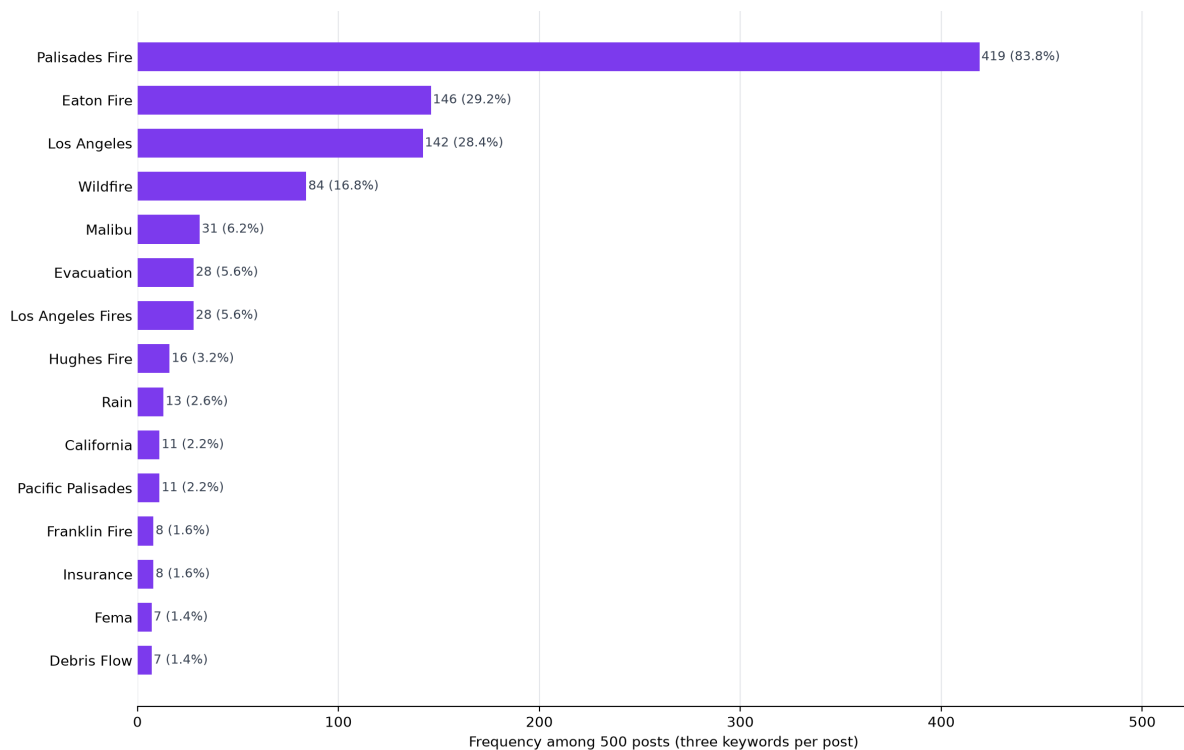


Figure 7. The 15 most frequent normalized LLM keywords.

5.4 Prominent and unexpected themes

Rank	Keyword	Frequency	Posts
1	palisades fire	419	83.8%
2	eaton fire	146	29.2%
3	los angeles	142	28.4%
4	wildfire	84	16.8%
5	malibu	31	6.2%
6	evacuation	28	5.6%
7	los angeles fires	28	5.6%
8	hughes fire	16	3.2%
9	rain	13	2.6%
10	california	11	2.2%
11	pacific palisades	11	2.2%
12	franklin fire	8	1.6%

The expected focal-event and Los Angeles terms dominate, but related-event and recovery topics also appear. In particular, Eaton Fire references show that users often discussed the January 2025 Los Angeles fires together. Weather, debris-flow, burn-scar, insurance, legal, and rebuilding terms demonstrate that the conversation extended beyond active flames into secondary hazards and long-term recovery. A sparse, unexpected tail including Will Rogers, Volkswagen, and Palestine reveals cross-topic hashtag reuse or incidental references; these terms were retained rather than manually removed.

6. Limitations

- Mastodon is decentralized; mastodon.social returns only posts known to that instance.
- Hashtag collection overrepresents users who tag posts and does not capture all relevant discussion.
- Reply context was capped, so some cascades may remain incomplete.
- The 200-user graph is a relevance-focused sample expanded from six seeds, not the complete social graph.
- Centrality measures are sensitive to sampling and edge definitions.
- A 3B-parameter LLM can simplify or inconsistently normalize themes; structured output and transparent post-level results make auditing possible.

7. Reproducibility and Run Instructions

Create the virtual environment, install dependencies, add the Mastodon token only to `.env`, and run the pipeline below. The real token is excluded from Git. Ollama and the Llama 3.2 model are required only for the LLM step.

```
python -m venv .venv
.\venv\Scripts\Activate.ps1
python -m pip install -r requirements.txt
python src/test_connection.py
python src/collect_posts.py
python src/collect_users.py
python src/build_networks.py
python src/analyze_networks.py
python src/plot_user_measures.py
ollama pull llama3.2:3b
python src/extract_keywords_llm.py
python src/analyze_content.py
python src/generate_report.py
```

8. References

- [1] CAL FIRE. *Palisades Fire incident record*. <https://www.fire.ca.gov/incidents/2025/1/7/palisades-fire>
- [2] Mastodon Documentation. *API introduction and timelines*. <https://docs.joinmastodon.org/client/intro/>
- [3] NetworkX Developers. *NetworkX documentation*. <https://networkx.org/documentation/stable/>
- [4] Gephi. *The Open Graph Viz Platform*. <https://gephi.org/>
- [5] Ollama. *Structured Outputs*. <https://docs.ollama.com/capabilities/structured-outputs>
- [6] Ollama Model Library. *Llama 3.2*. <https://ollama.com/library/llama3.2>
- [7] Mueller, A. *word_cloud documentation*. https://amueller.github.io/word_cloud/
- [8] Bastian, M., Heymann, S., and Jacomy, M. (2009). *Gephi: An Open Source Software for Exploring and Manipulating Networks*. ICWSM.